

# Lecture 32

$$\textcircled{1} \quad \frac{du}{dt} = A u = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} u$$

$$i) \quad u_t = c_1 e^{\lambda_1 t} x_1 + c_2 e^{\lambda_2 t} x_2 + \dots$$

→ singular matrix so,  $\lambda_1 = 0$

$$A - \lambda I = \begin{bmatrix} -\lambda & -1 & 0 \\ 1 & -\lambda & -1 \\ 0 & 1 & -\lambda \end{bmatrix}$$

$$-\lambda (\lambda^2 + 1) + 1 (-\lambda) = 0$$

$$\lambda (\lambda^2 + 2) = 0$$

$$\lambda = \pm \sqrt{2} i$$

Next fact  $\rightarrow$  Skew-symmetric matrices have pure imaginary eigenvalues.

$$\lambda = 0, \sqrt{2}i, -\sqrt{2}i$$

$$u_t = c_1 x_1 + c_2 e^{(\sqrt{2}i)t} x_2 + c_3 e^{(-\sqrt{2}i)t} x_3$$

ii) When does the solution return to its initial value?

$\rightarrow$  We have to find the  $t$  for which

$$u_t = u_0 = c_1 x_1 + c_2 x_2 + c_3 x_3$$
$$e^{(\sqrt{2}i)t} = e^{-(\sqrt{2}i)t} = 1$$

$$e^{\sqrt{2}it} = \cos(\sqrt{2}t) + i \sin(\sqrt{2}t)$$

$$e^{-\sqrt{2}it} = \cos(\underline{-\sqrt{2}t}) + i \sin(-\sqrt{2}t)$$

$$\Rightarrow 1 \text{ at } t=0$$

$$\sqrt{2}t = 2\pi k$$

$$\boxed{t = \sqrt{2} k \pi}$$

Another neat fact

The eigenvectors of both symmetric & skew symmetric matrices are orthogonal.

Condition for orthogonal eigenvectors  $\rightarrow$

$$\boxed{A A^T = A^T A} \rightarrow \text{symmetric / skew-sym / orthogonal}$$

iii) Find  $e^{At}$

$$u_t = e^{At} u_0$$

→ If  $A$  can be diagonalized

$$e^{At} = S e^{\Lambda t} S^{-1}$$

↓

$$\begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{bmatrix}$$

②  $A_{3 \times 3}$

$$\lambda_1 = 0$$

$$\lambda_2 = c$$

$$\lambda_3 = 2$$

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$x_3 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

i) Is the matrix diagonalizable?

Yes, independent eigenvectors

it works for all  $c$ 's.

ii) For which values of  $c$  is  $A$  symmetric?

All real  $c$ 's as  $\lambda$ 's are orthogonal.

iii) When will it be positive definite?

Never as  $\lambda_1 = 0$

can be semi positive definite  
at best when  $c \geq 0$

iv) Is it a markov matrix?

→ For markov matrices  $\lambda \leq 1$   
and need to have at least  
one  $\lambda = 1$

Since  $\lambda = 2$ , it cannot be  
a markov matrix.

v) could  $\frac{A}{2}$  be a projection matrix?

$$\left(\frac{A}{2}\right)^2 = \frac{1}{4} A^2 = \frac{1}{4} S \Lambda^2 S^{-1} = S \left(\frac{\Lambda}{2}\right)^2 S^{-1}$$

$$\begin{bmatrix} 0 & 1 \\ \frac{c}{2} & 1 \end{bmatrix} > \begin{bmatrix} 0 & 1 \\ \frac{c}{2} & 1 \end{bmatrix}$$

$$\frac{c^2}{4} - \frac{c}{2} = 0$$

$$c(c-2) = 0$$

$$c = 0, 2$$

$$\textcircled{3} \quad A = (\text{ortho}) (\text{diag}) (\text{ortho})$$

$$= U \Sigma V^T$$

$$\sim A^T A = V (\Sigma^T \Sigma) V^T = \underline{V \Sigma^2 V^T}$$

$$\begin{array}{c} A A^T = U \Sigma^2 U^T \\ - \quad - \quad - \quad - \quad - \end{array}$$

For symmetric matrices,

$V = U =$  eigenvector matrix.

$\Rightarrow$  To choose the sign on the eigenvectors we use this rule  $\rightarrow$

$$A v_i = \sigma_i u_i$$



$$9) \quad \Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$$

$$u = \begin{bmatrix} u_1 & u_2 \end{bmatrix}$$

$$v = \begin{bmatrix} v_1 & v_2 \end{bmatrix}$$

Find a vector in nullspace of  $A$

$$A \Rightarrow \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -v_1 \\ -v_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \cdot x \\ v_2 \cdot x \end{bmatrix}$$

$$\begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} 3 \cdot v_1 \cdot x \\ 0 \end{bmatrix}$$

$$v_1 \cdot x = 0$$

We know that  $v_1 \cdot v_2 = 0$   $\rightarrow$   $v_1$  is the nullspace

④  $A \Rightarrow$  symmetric & orthogonal.

$$\cdot A^T = A = A^{-1}$$

①  $\lambda$ 's can be  $\rightarrow$  real, exist in pair of inverse  
 $\left[ \frac{1}{\lambda}, \lambda \right]$

$$\Rightarrow Qx = \lambda x$$

Take length of both sides.

$$\Rightarrow \|Qx\| = |\lambda| \|x\|$$

$\|Qx\| = \|x\|$  as orthogonal matrices don't change magnitude.

$\downarrow$   
next fact

$$\boxed{|\lambda| = 1}$$

So, the eigenvalues can be 1, -1

② A is sure to be symmetric?

Nope.

③ No repeated eigenvalues?

False, can have repeated eigenvalues

④ Is it diagonalizable?

$$U \Sigma^2 U^T = V \Sigma^2 V^T = AA^T = A^T A = A^2$$

So,  $V = U$  and yes it is  
diagonalizable.

⑤ Is it non singular?

Yes, the  $\lambda$ 's can never be 0.

⑥ Show that  $\frac{1}{2}(A+I)$  is a projection matrix.

$$\Rightarrow A^T = A = A^{-1}$$

$$\frac{1}{4}(A+I)^2 = (A^2 + I^2 + 2A) \frac{1}{4}$$

$$A^2 = A \cdot A^T = I$$

$$(2 \cdot I + 2A) \frac{1}{4} = \underline{\underline{\frac{1}{2}(A+I)}}$$

Thus,  $\left(\frac{1}{2}(A+I)\right)^2 = \frac{1}{2}(A+I)$   
projection confirmed.

eigen values of  $\frac{1}{2}(A+I)$

$$\Rightarrow (A+I) \rightarrow 2, 0$$
$$\frac{1}{2}(A+I) = 1, 0$$

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